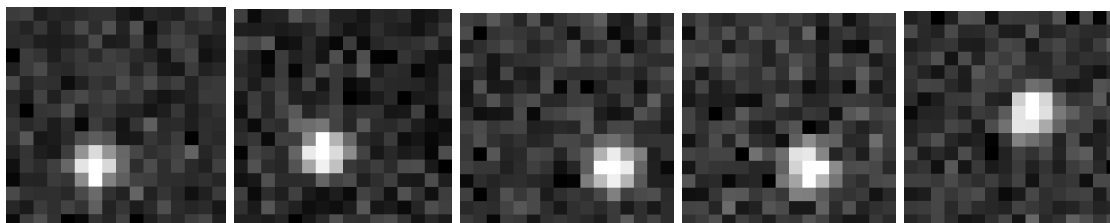


Multiple Emitters SMLM-KAN Localization Report (M=1 -10)

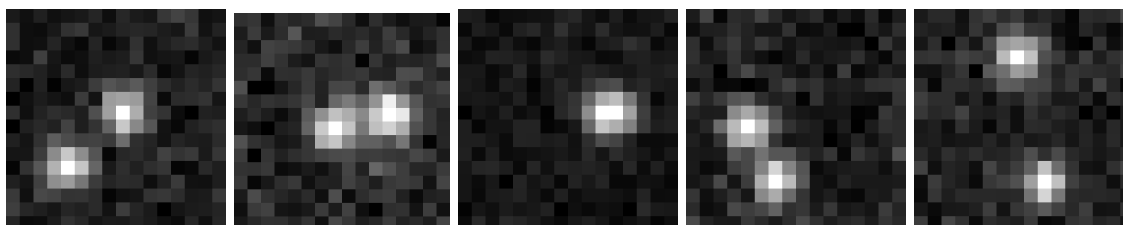
Part 1 — Data and Training

Dataframes samples:

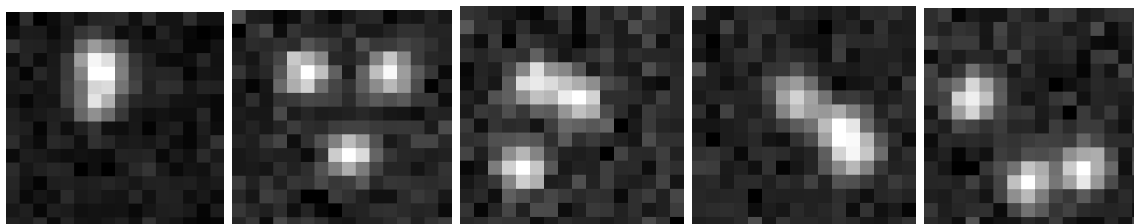
M=1



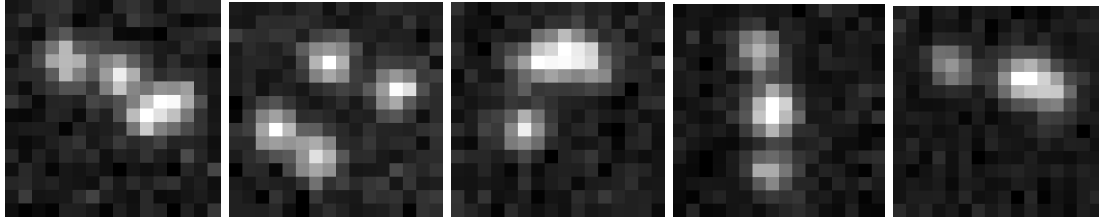
M=2



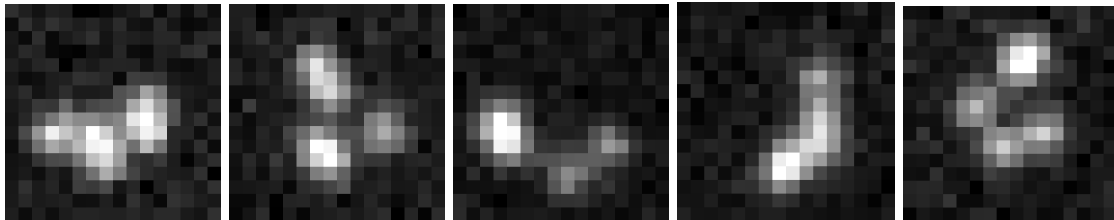
M=3



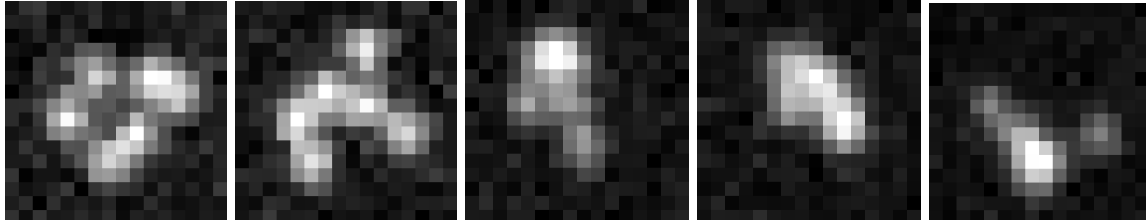
M=4



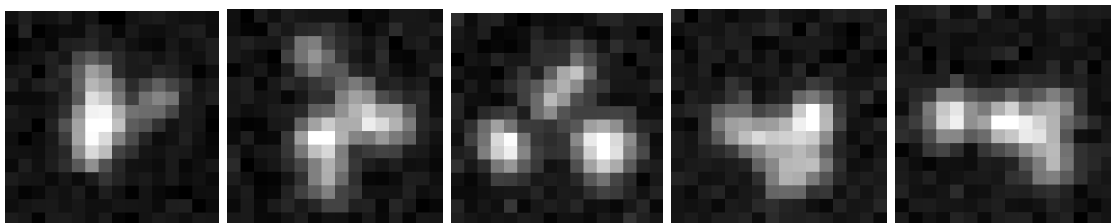
M=5



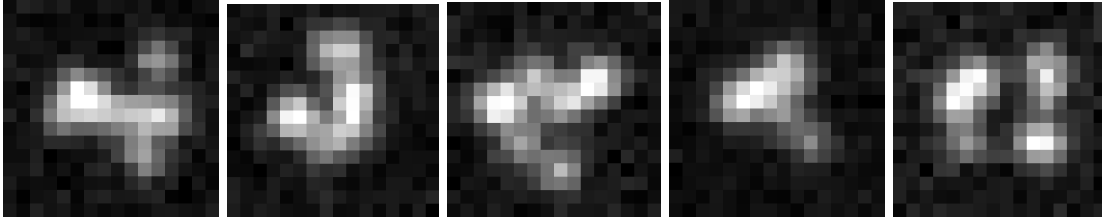
M= 6



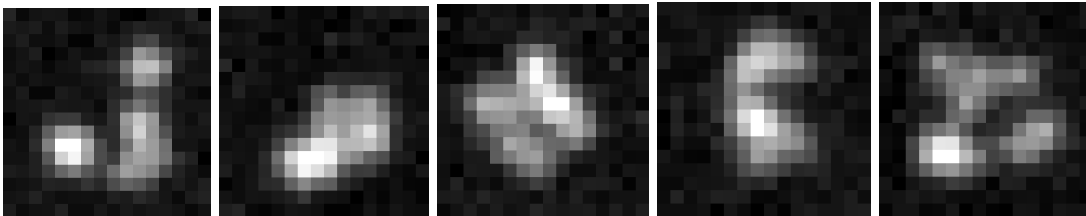
M= 7



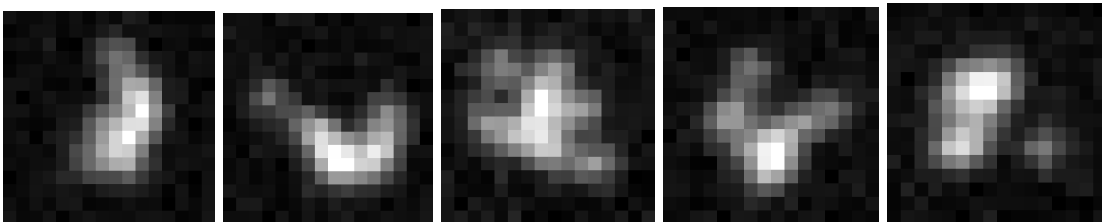
M= 8



M= 9



M= 10



Dataset split sizes

Train/validation split is 90% / 10% of total frames per M. Blind test set: 500 frames in combined_test/ (50 frames per M).

M	Train	Validation	Test (combined)
1	1000	100	50
2	20,000	200	50
3	50,000	500	50
4	100,000	1,000	50
5	200,000	2,000	50
6	500,000	5,000	50
7	1,000,000	10,000	50
8	2,000,000	20,000	50
9	3,000,000	30,000	50
10	5,000,000	50,000	50
Total	11,871,000	118,800	500

GPU and training time (per-M KAN models)

GPU: NVIDIA GeForce RTX 4060 Laptop GPU (CUDA, mixed precision)

Training time in total 2 hours.

M	Epochs run	Batch size	Wall time (min)
1	100	512	0.2
2	150	512	11.0
3	100	512	17.1
4	50	512	25.5
5	20	512	1.9
6	20	256	4.5
7	12	256	5.8
8	12	256	10.0
9	8	128	10.7
10	8	128	13.3

Part 2 — Network Architecture

Pipeline: CNN M-classifier → KAN localizer → RMSMD.

Fixed-M KAN localizer (FixedMKANLocalizer)

Input: 16×16×1 frame

SpatialEncoder (CNN):

- $M \leq 5$ (shallow): Conv32 → Conv64 → MaxPool → Conv64 → AdaptiveAvgPool 4×4 → 1024-d features
- $M \geq 6$ (deep): Conv32 → Conv64 → Conv128 → MaxPool → Conv128 → AdaptiveAvgPool 4×4 → 2048-d features

FastKAN backbone (Gaussian-spline KAN layers with ReLU):

- $M \leq 2$: width [in, 128, 128], grid size = 5
- $M 3-5$: width [in, 384, 192, 128], grid size = 5
- $M 6-8$: width [in, 512, 256, 128], grid size = 7
- $M 9-10$: width [in, 768, 384, 192], grid size = 7

M parallel heads (one per emitter slot):

- Classification head: Linear → softmax over 80 candidate centers

- Offset head: $\text{Linear}(2) \rightarrow \tanh \times \text{offset_scale_nm}(M)$

Decoded position: $\sum(\text{probability} \times \text{candidate coordinate}) + \text{offset} \rightarrow (x, y)$ in nm

Training loss: cross-entropy + smooth L1 coordinate loss (pixel units), CE warmup, AdamW, cosine LR, AMP on GPU

M-count classifier (MCountClassifier)

$16 \times 16 \times 1 \rightarrow \text{Conv32} \rightarrow \text{Conv64} \rightarrow \text{MaxPool} \rightarrow \text{Conv128} \rightarrow \text{AdaptiveAvgPool } 2 \times 2$

$\rightarrow \text{Flatten} \rightarrow \text{Linear}(512 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 10) \rightarrow \text{argmax} + 1$

Predicts $M \in \{1, \dots, 10\}$ to select the correct KAN checkpoint at inference.

Part 3 — RMSMD Results and Results on dataframes

M	RMSMD (nm)
1	9.2
2	17.7
3	24.7
4	38.9
5	45.6
6	57.5
7	72.6
8	74.1
9	87.4
10	102.3

GT - green dot

Predicted - red dot

